

Sorting algorithms & Divide and Conquer

- Elementary Sorting Algorithms
- Merge Sort
- Divide and Conquer

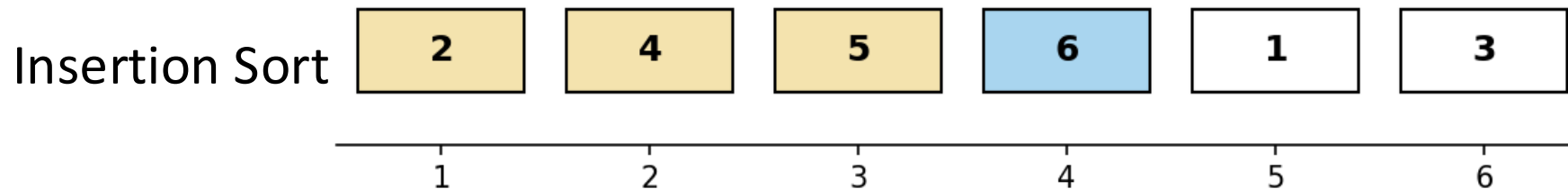
Yanlin Zhang | DSAA 2043 Spring 2026

Elementary Sorting Algorithms (Recap.)

The Problem of Sorting

- Input
 - A sequence of n numbers $\langle a_1, a_2, \dots, a_n \rangle$
- Output
 - Permutation $\langle a'_1, a'_2, \dots, a'_n \rangle$ such that $a'_1 \leq a'_2 \leq \dots \leq a'_n$
- Example
 - Input: 8 2 4 9 3 6
 - Output: 2 3 4 6 8 9
- Sorting is a fundamental operation:
 - Searching (Binary Search requires sorted arrays)
 - Data Processing (Efficient indexing in databases)
 - Graph Algorithms (Kruskal's algorithm for MST)
 - Bioinformatics (Sorting datasets before analysis)

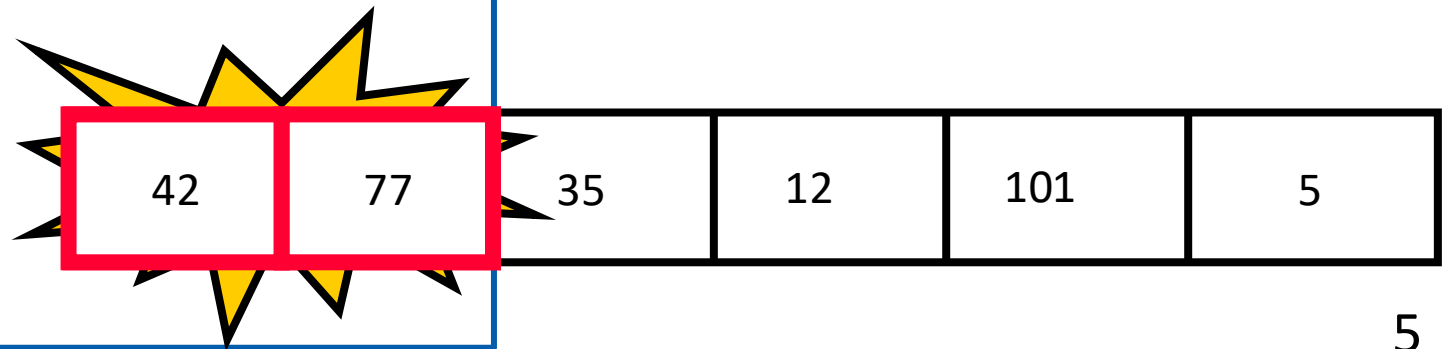
- Always consider the array has **two portions**: the **sorted portion** followed by the **unsorted portion**.
 - Initially, the sorted portion is empty, and the entire array is unsorted.
 - Enlarge the sorted portion by adding one more number in every iteration, and shrink the unsorted portion accordingly.



Bubble Sort (Recap.)

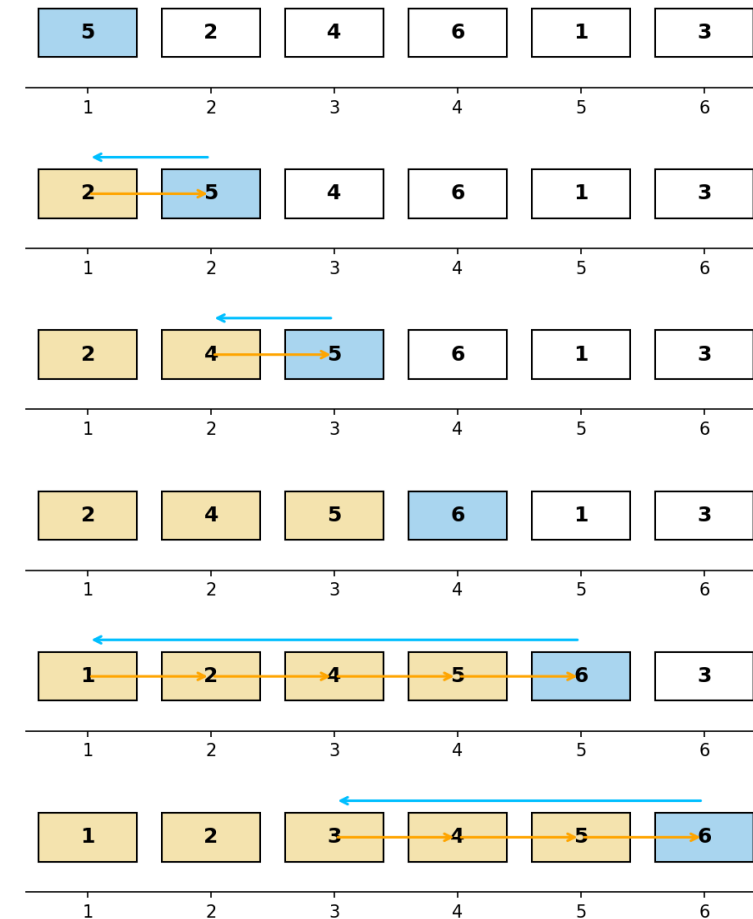
```
algorithm Bubblesort(A)
  to_do: index isoftype Num
  did_swap: Boolean
  to_do  $\leftarrow$  N - 1
  did_swap  $\leftarrow$  true

  while (to_do > 0) or (not did_swap):
    index  $\leftarrow$  1
    did_swap  $\leftarrow$  false
    while index < to_do + 1
      if(A[index] > A[index + 1]) then
        Swap(A[index], A[index + 1])
        did_swap  $\leftarrow$  true
      endif
      index  $\leftarrow$  index + 1
    end while
    to_do  $\leftarrow$  to_do - 1
  end while
```



Insertion Sort (Recap.)

Insertion Sort Step-by-Step



Algorithm InsertionSort(A)

Input: An array A with n elements

Output: A sorted in non-decreasing order

```
1. for i ← 2 to length(A) do
2.     key ← A[i] // Current element to be inserted
3.     j ← i - 1 // Start comparing from the previous element
4.     while j > 0 and A[j] > key do
5.         A[j + 1] ← A[j] // Shift element to the right
6.         j ← j - 1
7.     end while
8.     A[j + 1] ← key // Place key in the correct position
9. end for
10. return A // Sorted array
```

Selection sort (Recap.)

- Continuously finds the smallest element from the unsorted part and swaps it with the first unsorted position.

```
Input: An array A of size n (1-based index)
Output: A sorted array A in non-decreasing order
Algorithm SelectionSort(A, n):
    for i ← 1 to n do
        min_index ← i
        for j ← i+1 to n do
            if A[j] < A[min_index] then
                min_index ← j
        swap A[i] and A[min_index]
    return A
```

Pass	Unsorted Part	Min Element	Swap	Updated Array
1	[6, 3, 8, 5, 2]	2 (at index 5)	Swap A[1] ↔ A[5]	[2, 3, 8, 5, 6]
2	[3, 8, 5, 6]	3 (at index 2)	No swap needed	[2, 3, 8, 5, 6]
3	[8, 5, 6]	5 (at index 4)	Swap A[3] ↔ A[4]	[2, 3, 5, 8, 6]
4	[8, 6]	6 (at index 5)	Swap A[4] ↔ A[5]	[2, 3, 5, 6, 8]
5	[8]	No need to swap	Done	[2, 3, 5, 6, 8]

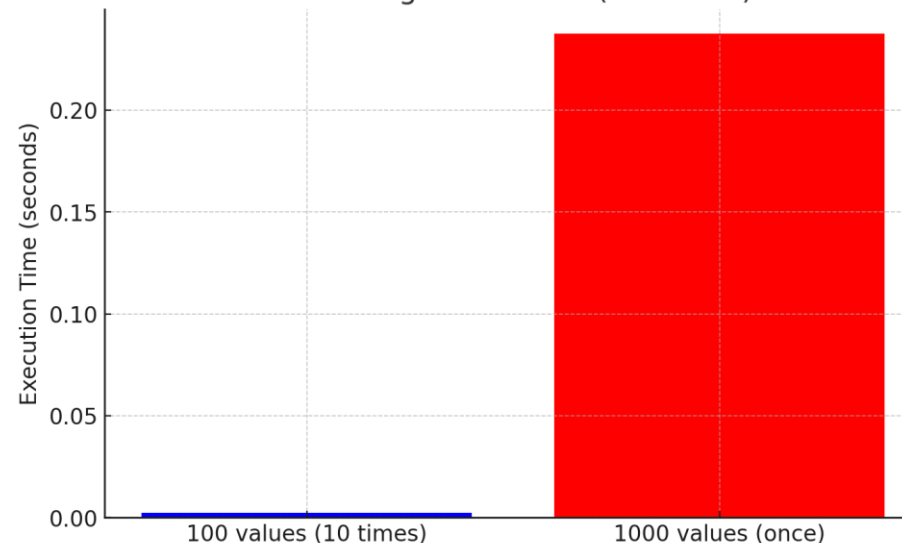
Merge Sort

The Problem with $O(n^2)$ Sorting Algorithms

Algorithm	Comparisons (Worst Case)	Swaps (Worst Case)
Bubble Sort	$O(n^2)$	$O(n^2)$
Selection Sort	$O(n^2)$	$O(n)$
Insertion Sort	$O(n^2)$	$O(n)$

- We need a sorting method that **reduces swaps and comparisons**.

Bubble Sort Performance: Sorting 100 Values (10 Times) vs. 1000 Values (Once)



How Would You Sort 1000 Pieces of Paper?

- **Method A:** Uses Insertion Sort strategy (picking one paper at a time and inserting it in order).
- **Method B:** Uses Merge-like strategy (first sorts small sections, then merges them together).
 - How many comparisons are needed for merging two sorted sections?
- **Result:** Method B finishes much faster!
- **Lesson:** Sorting small sections first, then merging, is faster than moving elements one by one.
 - Fewer swaps and comparisons = faster sorting!
 - Merging sorted sections is easier than sorting everything from scratch.

Algorithm Merge-Sort $A[1 \dots n]$

1. If $n = 1$, done
2. Recursively sort $A[1 \dots \lceil n/2 \rceil]$
and $A[\lceil n/2 \rceil + 1 \dots n]$
3. “*Merge*” the 2 sorted lists

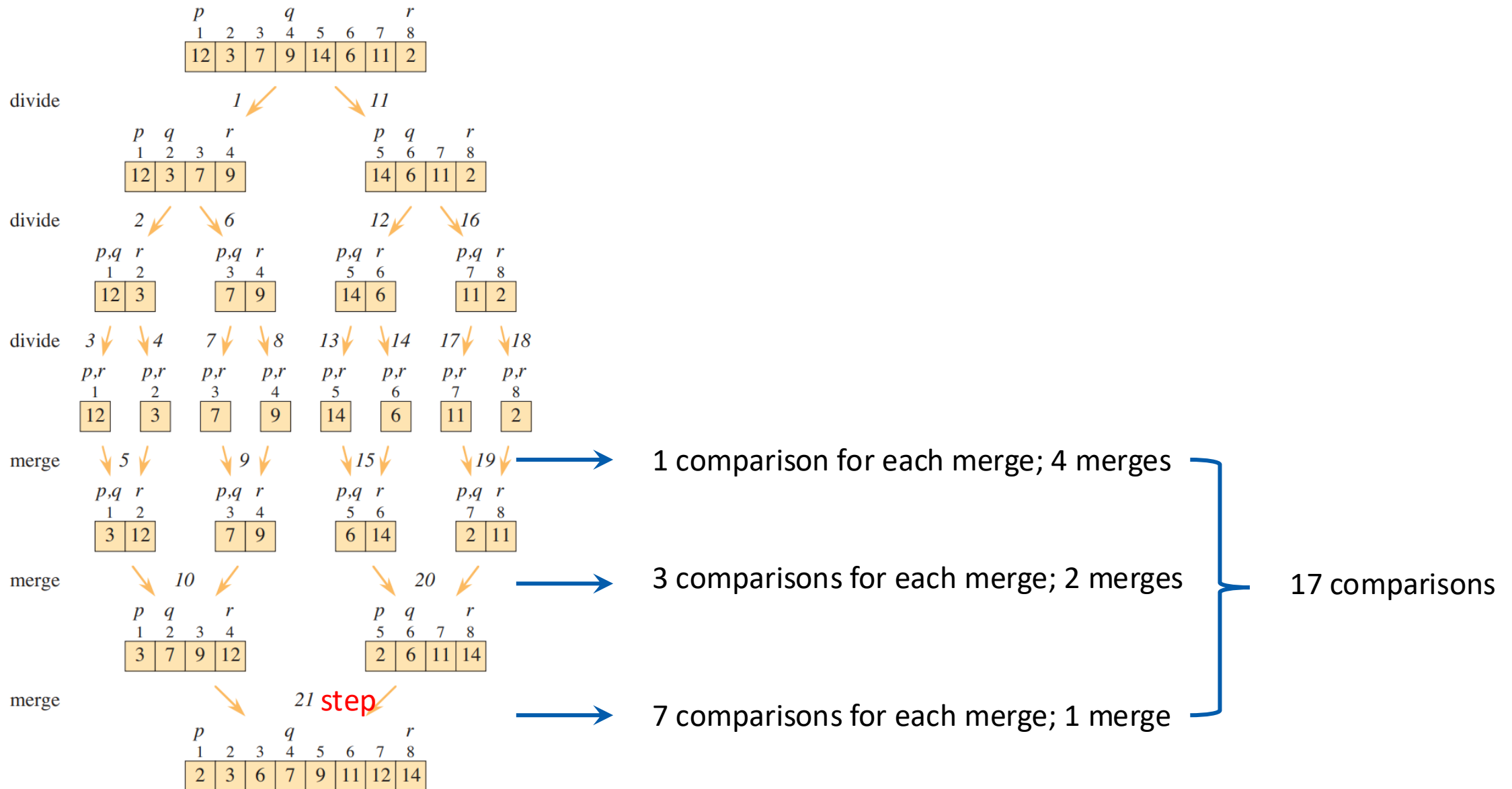
Key subroutine: **MERGE**

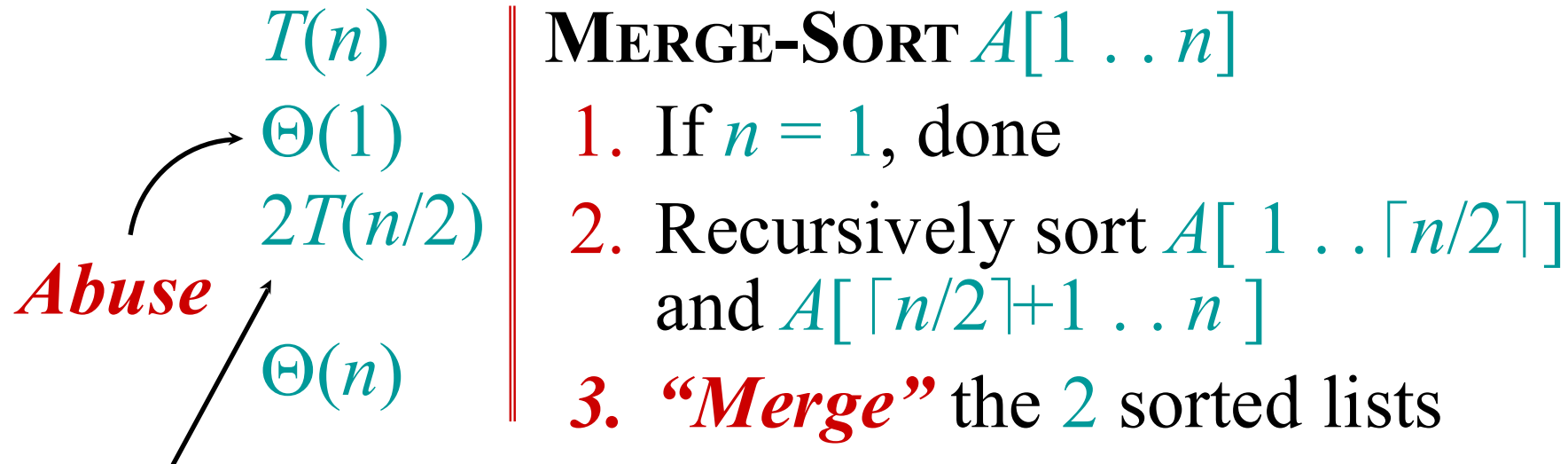
Merging Two Sorted Arrays (Algo. Merge)

20	12
13	11
7	9
2	1

Time = $\Theta(n)$ to merge a total
of n elements (linear time)

Merge-Sort Example





Sloppiness: Should be $T(\lceil n/2 \rceil) + T(\lfloor n/2 \rfloor)$,
but it turns out not to matter asymptotically.

$$T(n) = \begin{cases} \Theta(1) & \text{if } n = 1; \\ 2T(n/2) + \Theta(n) & \text{if } n > 1. \end{cases}$$

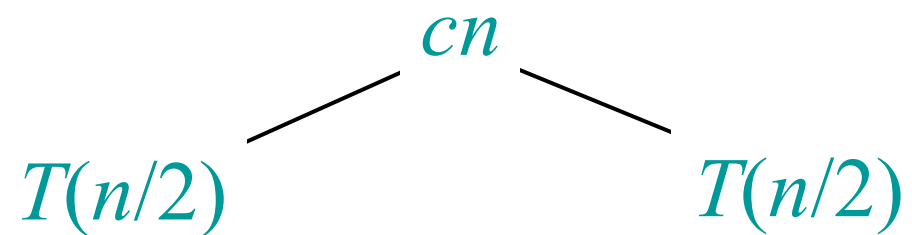
- We shall usually omit stating the base case when $T(n) = \Theta(1)$ for sufficiently small n , but only when it has no effect on the asymptotic solution to the recurrence.
- CLRS provides several ways to find a good upper bound on $T(n)$.

Solve $T(n) = 2T(n/2) + cn$, where $c > 0$ is constant.

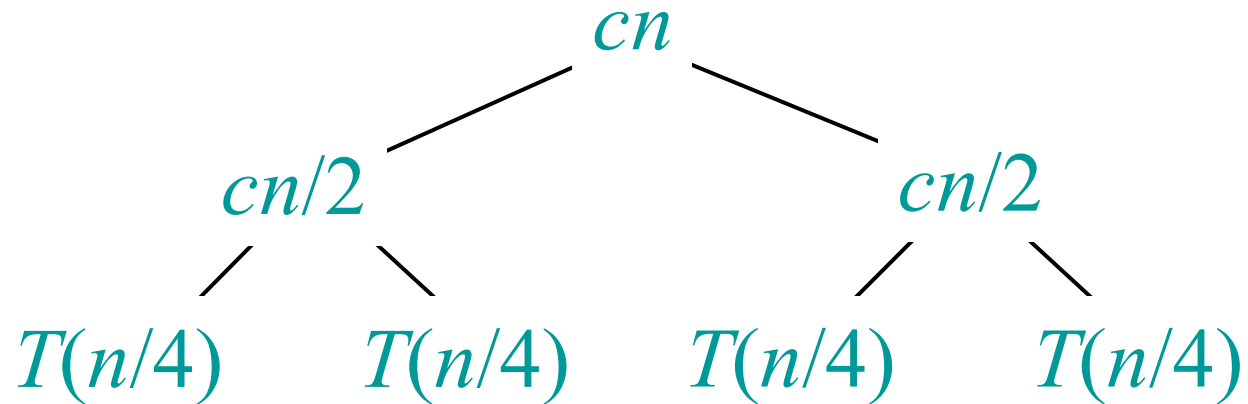
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$$T(n)$$

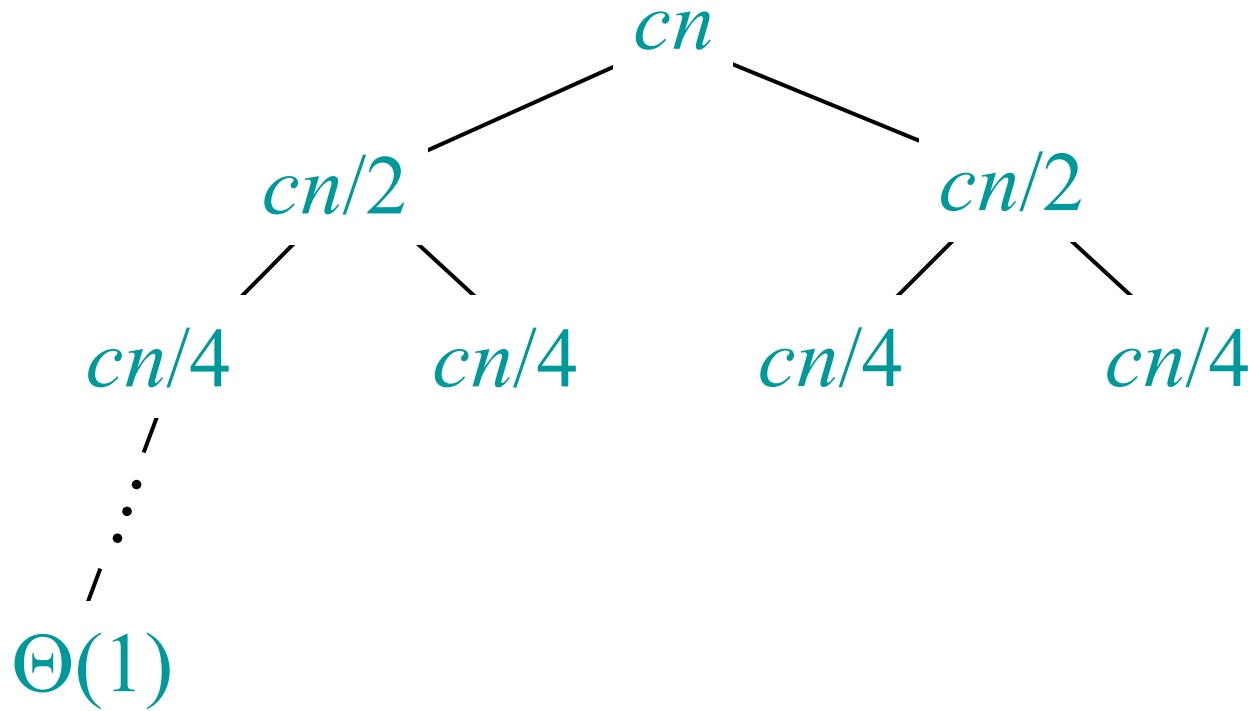
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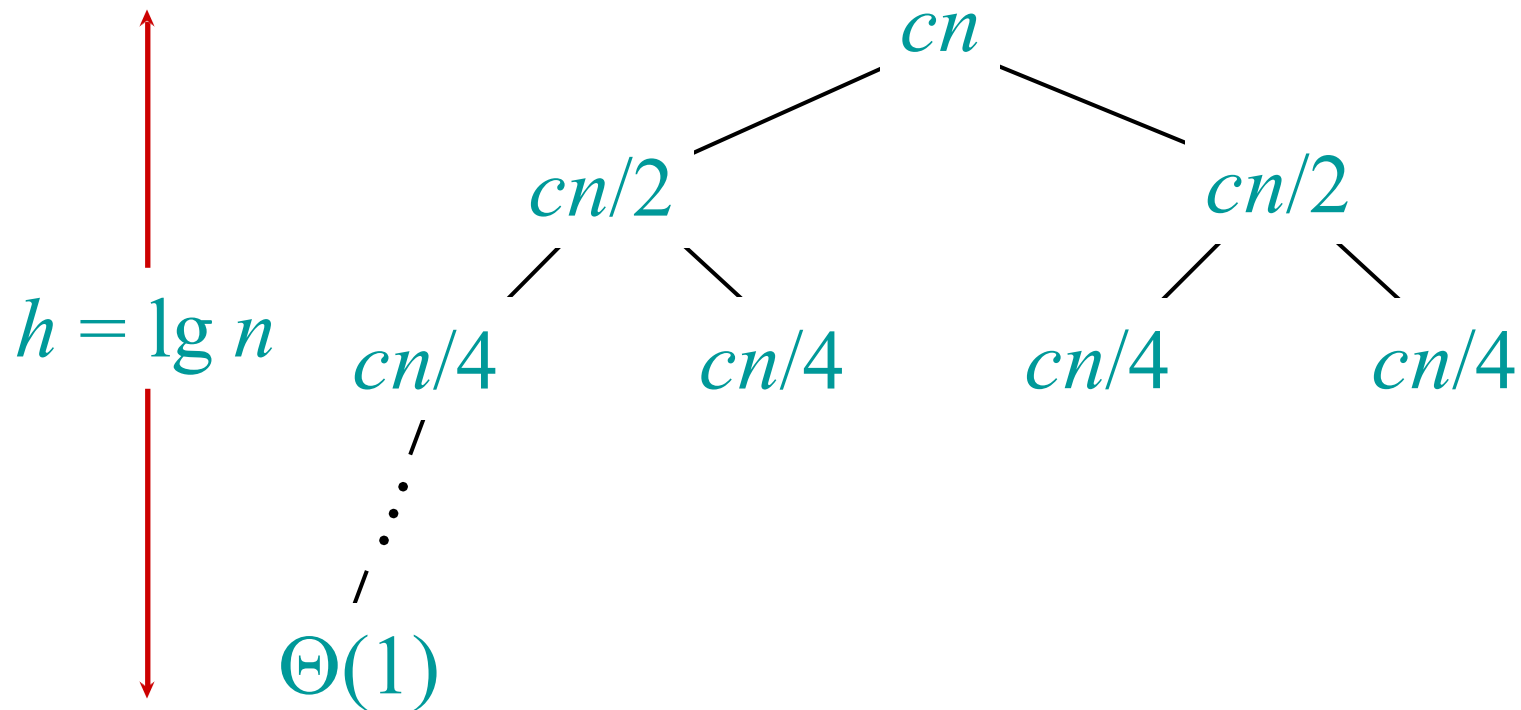
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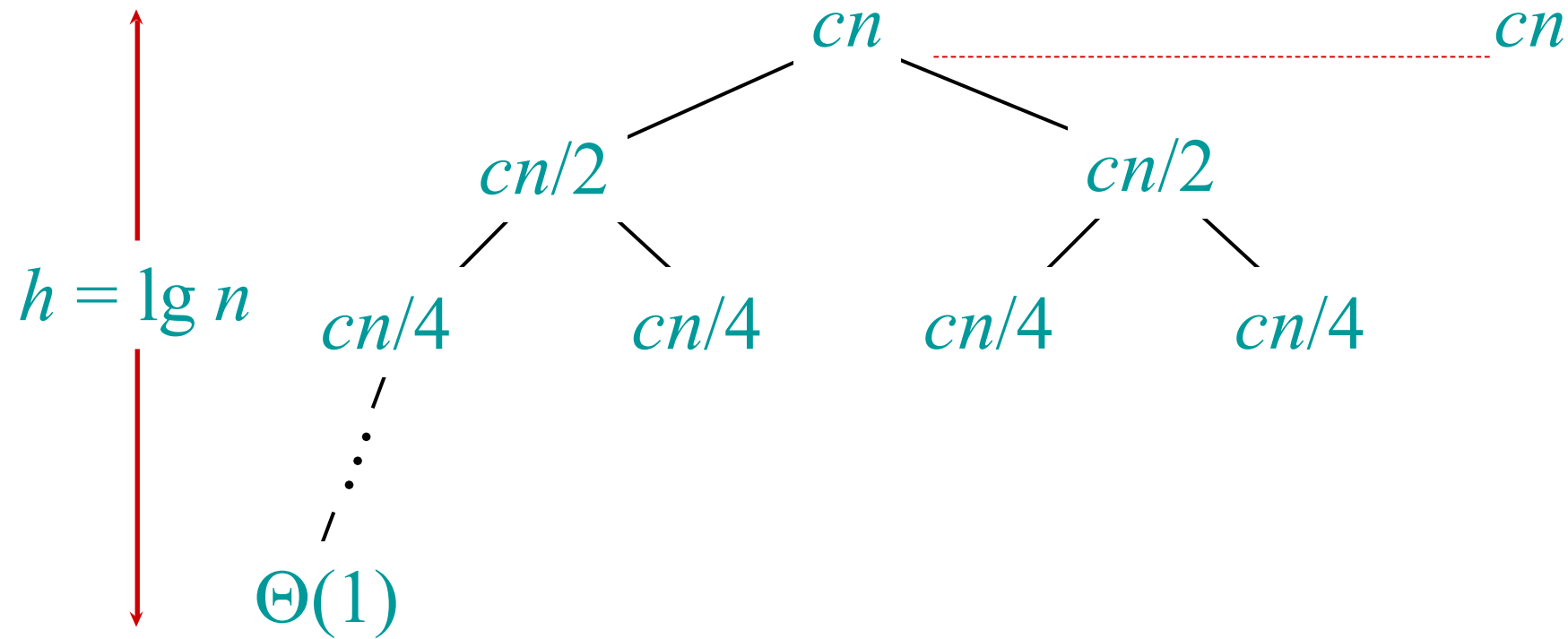


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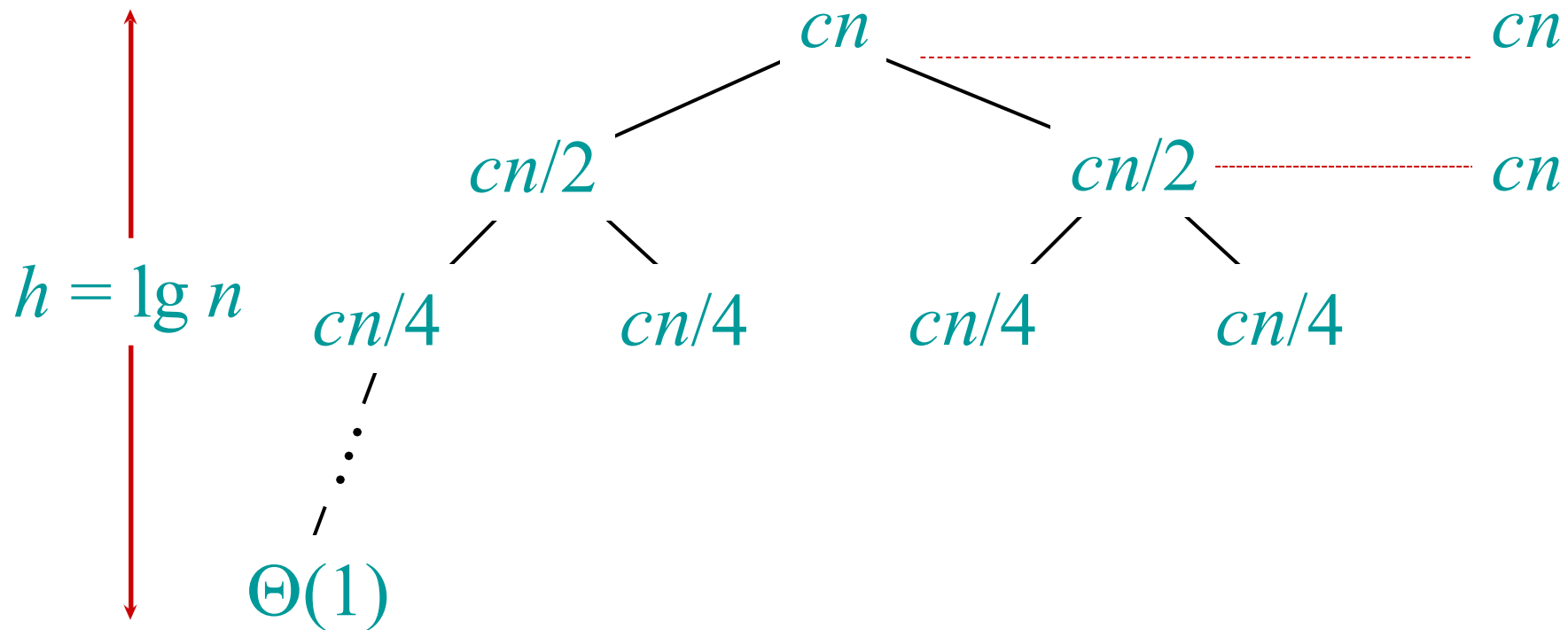


Recursion Tree

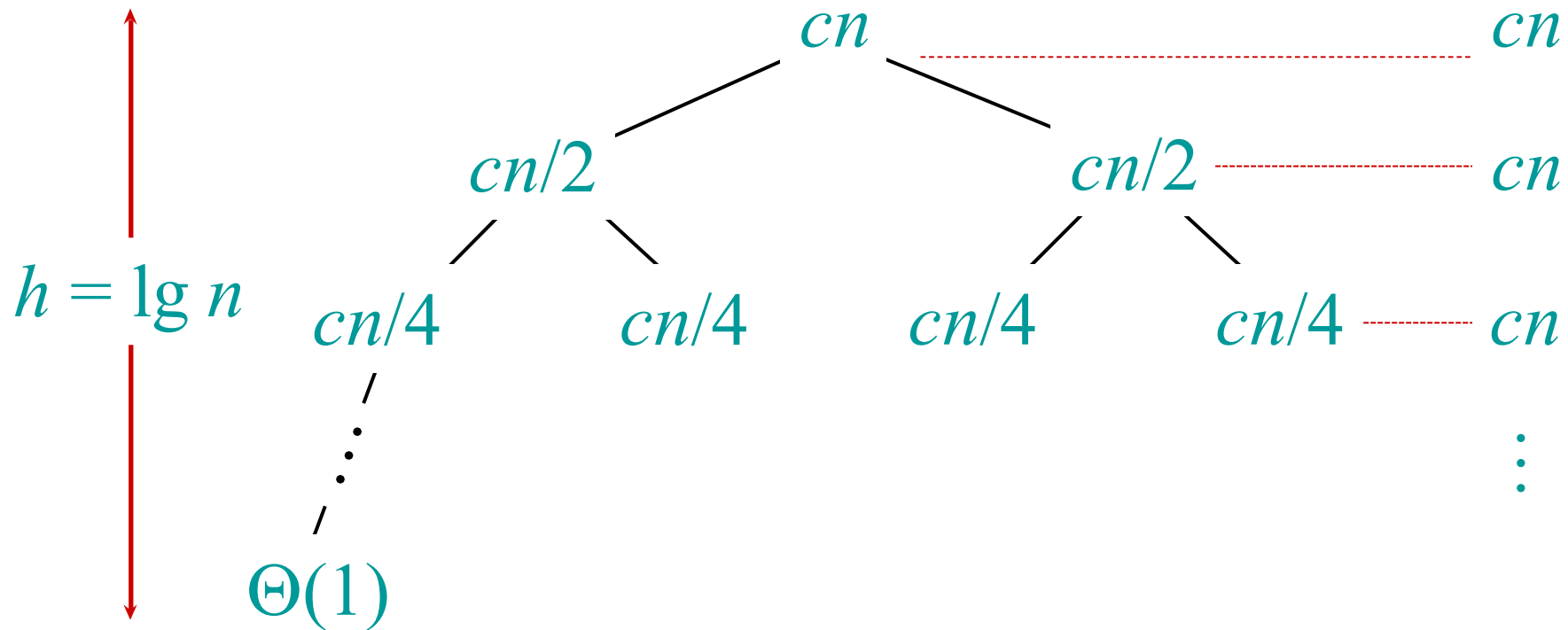
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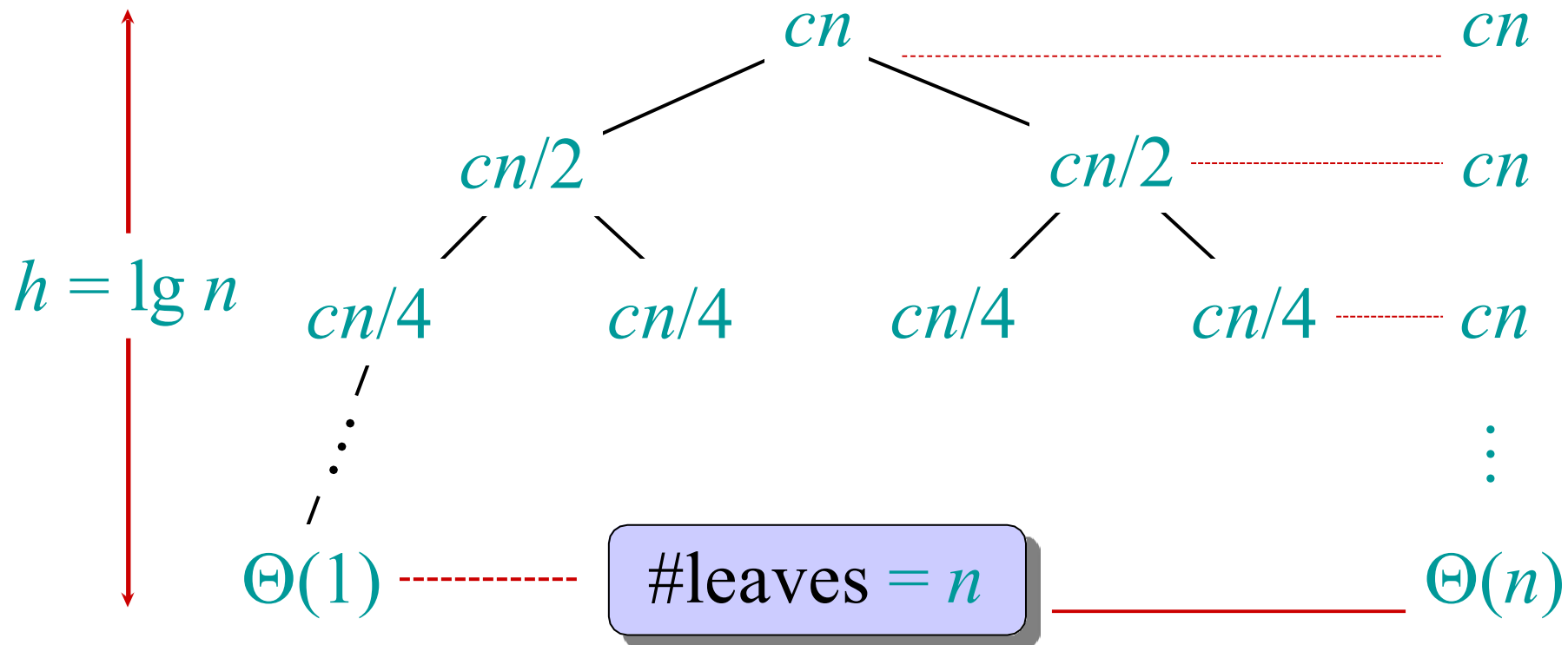


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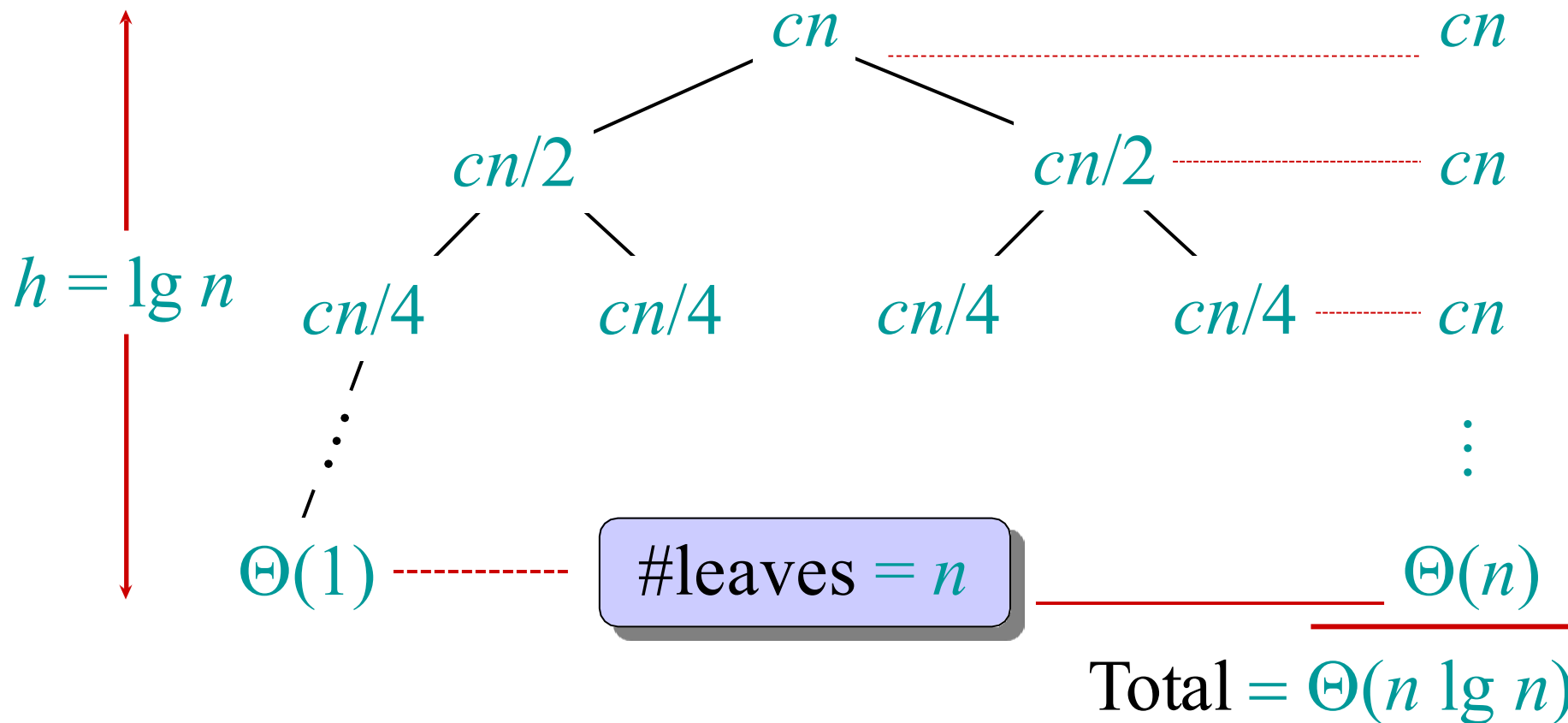
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Recursion Tree

Solve $T(n) = 2T(n/2) + cn$, where $c > 0$ is constant.



- $\Theta(n \lg n)$ grows more slowly than $\Theta(n^2)$.
- Therefore, merge sort asymptotically beats insertion sort in the worst case.
- In practice, merge sort beats insertion sort for $n > 30$ or so.
- Go test it out for yourself!

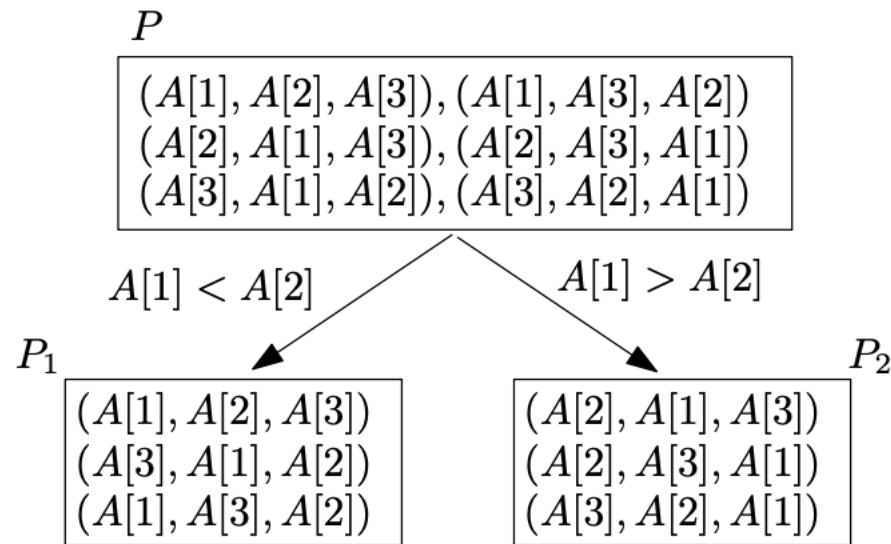
- Next, we will prove that Merge-Sort is optimal for **comparison-based algorithms**. In other words, every such algorithm must incur $\Omega(n \log n)$ time on at least one input.
- There are **$n!$** different ways to permute the n elements in the input array **A** .
- Example: For $n = 3$, 6 permutations:
 $A[1], A[2], A[3]$
 $A[1], A[3], A[2]$
 $A[2], A[1], A[3]$
 $A[2], A[3], A[1]$
 $A[3], A[1], A[2]$
 $A[3], A[2], A[1]$
- The goal of sorting is essentially to decide which of $n!$ Permutations is the final sorted order.

Formally, such an algorithm works by continuously shrinking a pool P of possible permutations.

- At the beginning, P contains all the $n!$ permutations.
- Every comparison allows the algorithm to discard all those permutations in P that are inconsistent with the comparison's result.
- Eventually, P has only 1 permutation left, which is thus the final sorted order.

In other words, at any moment, all the permutations that remain in P are possible results. The algorithm cannot terminate as long as $|P| \geq 2$.

- Shrinking the Pool: An Example



- In general, each comparison allows us to shrink P to either P_1 or P_2 .

Comparison-Based Sorting Lower Bound

Framework

1. $P \leftarrow$ all the $n!$ permutations of A
2. **while** $|P| > 1$
3. make a comparison between elements e_1 and e_2
4. **if** $e_1 < e_2$ **then**
5. $P \leftarrow P_1$, where P_1 is the set of permutations in P
 consistent with $e_1 < e_2$
6. **else**
7. $P \leftarrow P_2$, where P_2 is the set of permutations in P
 consistent with $e_1 > e_2$
8. **return** the permutation in P

Various algorithms differ in how they implement Step 3.

A Worst-Case Lower Bound

Formally, such an algorithm works by continuously shrinking a pool P of possible permutations.

- Note that one of P_1 and P_2 contains at least half of the permutations in P (i.e., either $|P_1| \geq |P|/2$ or $|P_2| \geq |P|/2$).
- The worst case happens when P always shrinks to the **larger** set between P_1 and P_2 .
- In this case, the size of P shrinks by at most half after each comparison.
- Hence, the number of comparisons required before $|P|$ decreases to 1 is $\log_2(n!)$.

The next slide shows $\log_2(n!) = \Omega(n \log n)$.

A Worst-Case Lower Bound

$$\begin{aligned}\log_2(n!) &= \sum_{i=1}^n \log_2 i \\ &\geq \sum_{i=n/2}^n \log_2 i \\ &\geq (n/2) \log_2(n/2) \\ &= \Omega(n \log n)\end{aligned}$$

We now conclude that any comparison-based algorithm must incur $\Omega(n \log n)$ time sorting n elements in the worst case.

Divide-and-Conquer

The Divide-and-Conquer Design Paradigm

1. *Divide* the problem (instance) into subproblems.
2. *Conquer* the subproblems by solving them recursively.
3. *Combine* subproblem solutions.

- 1. Divide:* Trivial.
- 2. Conquer:* Recursively sort 2 subarrays.
- 3. Combine:* Linear-time merge.

1. **Divide:** Trivial.
2. **Conquer:** Recursively sort 2 subarrays.
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$$T(n) = 2T(n/2) + \Theta(n)$$

subproblems

subproblem size

work dividing and combining

Master Theorem (Reprise)

$$T(n) = a T(n/b) + f(n)$$

CASE 1: $f(n) = O(n^{\log_b a - \varepsilon})$, constant $\varepsilon > 0$

$$\Rightarrow T(n) = \Theta(n^{\log_b a})$$

CASE 2: $f(n) = \Theta(n^{\log_b a} \lg^l n)$, constant $l \geq 0$

$$\Rightarrow T(n) = \Theta(n^{\log_b a} \lg^{l+1} n)$$

CASE 3: $f(n) = \Omega(n^{\log_b a + \varepsilon})$, constant $\varepsilon > 0$,
and regularity condition $a f(n/b) \leq c f(n)$,
constant $c < 1$ for all sufficiently large n

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Merge sort: $a = 2, b = 2 \Rightarrow n^{\log_b a} = n^{\log_2 2} = n$

$$\Rightarrow \text{CASE 2 } (l = 0) \Rightarrow T(n) = \Theta(n \lg n)$$

Find an element in a sorted array:

- 1. Divide:*** Check middle element.
- 2. Conquer:*** Recursively search **1** subarray.
- 3. Combine:*** Trivial.

Master Theorem (Proof)

$$T(n) = a T(n/b) + f(n)$$

Try to solve case 2 at home!

$$T(n) = \Theta(n^{\log_b a}) + \sum_{i=0}^{k-1} a^i f\left(\frac{n}{b^i}\right), \longrightarrow g(n) = \sum_{i=0}^{k-1} a^i f\left(\frac{n}{b^i}\right)$$

$$n = b^k, k = \log_b n, a^k = a^{\log_b n} = n^{\log_b a}$$

For case1: $f(n) = O(n^{(\log_b a) - \epsilon}), \epsilon > 0$

$$\begin{aligned} \text{We have: } g(n) &= O\left(\sum_{i=0}^{k-1} a^i \left(\frac{n}{b^i}\right)^{(\log_b a) - \epsilon}\right) \\ &= O\left(n^{(\log_b a) - \epsilon} \sum_{i=0}^{k-1} \left(\frac{ab^\epsilon}{b^{\log_b a}}\right)^i\right) \\ &= O\left(n^{(\log_b a) - \epsilon} \sum_{i=0}^{k-1} (b^\epsilon)^i\right) \\ &= O\left(n^{(\log_b a) - \epsilon} \sum_{i=0}^{k-1} (b^\epsilon)^i\right) \\ &= O\left(n^{(\log_b a) - \epsilon} \frac{n^\epsilon - 1}{b^\epsilon - 1}\right) \\ &= O(n^{\log_b a}) \end{aligned}$$

For case3: $f(n) = \Omega(n^{(\log_b a) + \epsilon}), \epsilon > 0$

$$af\left(\frac{n}{b}\right) \leq cf(n), c < 1$$

We have: $af\left(\frac{n}{b^2}\right) \leq cf\left(\frac{n}{b}\right)$

$\vdots$

$$af\left(\frac{n}{b^i}\right) \leq cf\left(\frac{n}{b^{i-1}}\right)$$

Multiply both sides: $a^i f\left(\frac{n}{b^i}\right) \leq c^i f(n)$

$$\begin{aligned} g(n) &= \sum_{i=0}^{k-1} a^i f\left(\frac{n}{b^i}\right) \leq \sum_{i=0}^{k-1} c^i f(n) = f(n) \sum_{i=0}^{k-1} c^i \\ &\leq f(n) \frac{1}{1-c} = \Theta(f(n)) \end{aligned}$$

Find an element in a sorted array:

- 1. *Divide*:** Check middle element.
- 2. *Conquer*:** Recursively search 1 subarray.
- 3. *Combine*:** Trivial.

Example: Find 9

3 5 7 8 9 12 15

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Recurrence for Binary Search

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subproblems *subproblem size* *work dividing and combining*

Recurrence for Binary Search

$$T(n) = 1 T(n/2) + \Theta(1)$$

subproblems *subproblem size* *work dividing and combining*

$$n^{\log_b a} = n^{\log_2 1} = n^0 = 1 \Rightarrow \text{CASE 2 } (l=0)$$
$$\Rightarrow T(n) = \Theta(\lg n).$$

Powering a Number

Problem: Compute a^n , where $n \in \mathbf{N}$.

Naive algorithm: $\Theta(n)$.

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Naive algorithm: $\Theta(n)$.

Divide-and-conquer algorithm:

$$a^n = \begin{cases} a^{n/2} \cdot a^{n/2} & \text{if } n \text{ is even;} \\ a^{(n-1)/2} \cdot a^{(n-1)/2} \cdot a & \text{if } n \text{ is odd.} \end{cases}$$

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$$T(n) = T(n/2) + \Theta(1) \Rightarrow T(n) = \Theta(\lg n).$$

Recursive definition:

$$F_n = \begin{cases} 1 & \text{if } n = 0; \\ 1 & \text{if } n = 1; \\ F_{n-1} + F_{n-2} & \text{if } n \geq 2. \end{cases}$$

0 1 1 2 3 5 8 13 21 34 ...

Recursive definition:

$$F_n = \begin{cases} 0 & \text{if } n = 0; \\ 1 & \text{if } n = 1; \\ F_{n-1} + F_{n-2} & \text{if } n \geq 2. \end{cases}$$

0 1 1 2 3 5 8 13 21 34 ...

Naive recursive algorithm: $\Omega(\phi^n)$

(exponential time), where $\phi = (1 + \sqrt{5})/2$
is the *golden ratio*.

Bottom-up:

- Compute $F_0, F_1, F_2, \dots, F_n$ in order, forming each number by summing the two previous.
- Running time: $\Theta(n)$.

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- Compute $F_0, F_1, F_2, \dots, F_n$ in order, forming each number by summing the two previous.
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Naive recursive squaring:

$F_n = \phi^n / 5$ rounded to the nearest integer.

- Recursive squaring: $\Theta(\lg n)$ time.
- This method is unreliable, since floating-point arithmetic is prone to round-off errors.

Theorem:
$$\begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n$$

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Algorithm: Recursive squaring.

Time = $\Theta(\lg n)$.

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Algorithm: Recursive squaring.

Time = $\Theta(\lg n)$.

Proof of theorem. (Induction on n .)

Base ($n = 1$):
$$\begin{bmatrix} F_2 & F_1 \\ F_1 & F_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^1$$

Inductive step ($n \geq 2$):

$$\begin{aligned} \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix} &= \begin{bmatrix} F_n & F_{n-1} \\ F_{n-1} & F_{n-2} \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{n-1} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \end{aligned}$$

Or Equivalently

$$\begin{bmatrix} F_{n+1} & F_n \end{bmatrix} = \begin{bmatrix} F_n & F_{n-1} \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Matrix Multiplication

Input: $A = [a_{ij}], B = [b_{ij}].$
Output: $C = [c_{ij}] = A \cdot B.$ $\left. \vphantom{\begin{matrix} A \\ B \\ C \end{matrix}} \right\} i, j = 1, 2, \dots, n.$

$$\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix}$$

$$c_{ij} = \sum_{k=1}^n a_{ik} \cdot b_{kj}$$

```
for  $i \leftarrow 1$  to  $n$ 
  do for  $j \leftarrow 1$  to  $n$ 
    do  $c_{ij} \leftarrow 0$ 
      for  $k \leftarrow 1$  to  $n$ 
        do  $c_{ij} \leftarrow c_{ij} + a_{ik} \cdot b_{kj}$ 
```

```
for  $i \leftarrow 1$  to  $n$ 
  do for  $j \leftarrow 1$  to  $n$ 
    do  $c_{ij} \leftarrow 0$ 
      for  $k \leftarrow 1$  to  $n$ 
        do  $c_{ij} \leftarrow c_{ij} + a_{ik} \cdot b_{kj}$ 
```

Running time = $\Theta(n^3)$

IDEA:

$n \times n$ matrix = 2×2 matrix of $(n/2) \times (n/2)$ submatrices:

$$\begin{bmatrix} r & s \\ t & u \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$C = A \cdot B$$

$$r = ae + bg$$

$$s = af + bh$$

$$t = ce + dg$$

$$u = cf + dh$$

8 mults of $(n/2) \times (n/2)$ submatrices

4 adds of $(n/2) \times (n/2)$ submatrices

IDEA:

$n \times n$ matrix = 2×2 matrix of $(n/2) \times (n/2)$ submatrices:

$$\begin{bmatrix} r & s \\ t & u \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$\begin{array}{l} r = ae + bg \\ s = af + bh \\ t = ce + dh \\ u = cf + dg \end{array} \left\{ \begin{array}{l} \text{recursive} \\ 8 \text{ mults of } (n/2) \times (n/2) \text{ submatrices} \\ 4 \text{ adds of } (n/2) \times (n/2) \text{ submatrices} \end{array} \right.$$

$C = A \cdot B$

Analysis of D&C Algorithm

$$T(n) = 8T(n/2) + \Theta(n^2)$$

submatrices *submatrix size* *work adding submatrices*

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No better than the ordinary algorithm.

- Multiply 2×2 matrices with only 7 recursive mults.

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$$P_2 = (a + b) \cdot h$$

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$$P_4 = d \cdot (g - e)$$

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7 mults, 18 adds/subs.

Note: No reliance on commutativity of mult!

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$$\begin{aligned} r &= P_5 + P_4 - P_2 + P_6 \\ &= (a + d)(e + h) \\ &\quad + d(g - e) - (a + b)h \\ &\quad + (b - d)(g + h) \\ &= ae + ah + de + dh \\ &\quad + dg - de - ah - bh \\ &\quad + bg + bh - dg - dh \\ &= ae + bg \end{aligned}$$

- 1. *Divide:*** Partition A and B into $(n/2) \times (n/2)$ submatrices. Form terms to be multiplied using $+$ and $-$.
- 2. *Conquer:*** Perform 7 multiplications of $(n/2) \times (n/2)$ submatrices recursively.
- 3. *Combine:*** Form C using $+$ and $-$ on $(n/2) \times (n/2)$ submatrices.

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Best to date (of theoretical interest only): $\Theta(n^{2.376...})$.

- Divide and conquer is just one of several powerful techniques for algorithm design.
- Divide-and-conquer algorithms can be analyzed using recurrences and the master method (so practice this math).
- The divide-and-conquer strategy often leads to efficient algorithms.